

Sprint 5: Consultas con MongoDB

NIVEL 1:

1. Crea una BD en MongoDB utilizando los archivos adjuntos. Resuelve:
 - a) Muestra los 2 primeros comentarios que aparecen en la base de datos.

```
> _MONGOSH
> use app_cine
< switched to db app_cine
> db.comments.find().limit(2)
< [
  {
    _id: ObjectId('5a9427648b0beebe69579cc'),
    name: 'Andrea Le',
    email: 'andrea_le@fakegmail.com',
    movie_id: ObjectId('573a1390f29313caabcd418c'),
    text: 'Rem officiis eaque repellendus amet eos doloribus. Porro dolor voluptatum voluptates neque culpa molestias. Voluptate unde nulla temporibu
    date: 2012-03-26T23:20:16.000Z
  }
  {
    _id: ObjectId('5a9427648b0beebe69579cf'),
    name: 'Greg Powell',
    email: 'greg_powell@fakegmail.com',
    movie_id: ObjectId('573a1390f29313caabcd41b1'),
    text: 'Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Commodi nisi sit placeat rerum vero cupiditate neque. Dolorum nihil ve
    date: 1987-02-10T00:29:36.000Z
  }
]
app_cine>
```

- b) ¿Cuántos usuarios tenemos registrados?

```
> db.users.countDocuments()
< 185
app_cine>
```

- c) ¿Cuántos cines existen en el estado de California?

```
> db.theaters.countDocuments({ "location.address.state": "CA" })
< 169
app_cine>
```

- d) ¿Cuál fue el primer usuario en registrarse?

```
> db.users.find().sort({ _id: 1 }).limit(1)
< {
  _id: ObjectId('59b99db4cfa9a34dcd7885b6'),
  name: 'Ned Stark',
  email: 'sean_bean@gameofthron.es',
  password: '$2b$12$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNIJ9RJAqMUQ74cr1J1Vu'
}
```

e) ¿Cuántas películas de comedia existen en nuestra base de datos?

```
> db.movies.countDocuments({ genres: "Comedy" })
< 7024
```

- Muéstrame todos los documentos de las películas producidas en 1932, pero que el género sea drama o estén en francés.

```
> db.movies.find({
  year: 1932,
  $or: [
    { genres: "Drama" },
    { languages: "French" }
  ]
})
< {
  _id: ObjectId('573a1391f29313caabcd9458'),
  plot: 'A young artist draws a face at a canvas on his easel. Suddenly the mouth on the drawing comes into life and starts talking. The artist tri
  runtime: 55,
  rated: 'UNRATED',
  cast: [
    'Enrique Rivero',
    'Elizabeth Lee Miller',
    'Pauline Carton',
    'Odette Talazac'
  ],
  num_mflix_comments: 1,
  poster: 'https://m.media-amazon.com/images/M/MV5BYWY3ODE5ZWVtYjlmYi00NjA4LTk4ZWYtMzBhZDE5MjY0YTUxXkEyXkFqcGdeQXVyNzI4MDMyMTU0._V1_SV1000_SX677_AL
  title: 'The Blood of a Poet',
  lastupdated: '2015-09-16 13:13:05.537000000',
  languages: [
    'French'
  ],
  released: 2010-05-20T00:00:00.000Z,
  directors: [
    'Jean Cocteau'
  ],
  writers: [
    'Jean Cocteau'
  ]
}
```

```
{
  _id: ObjectId('573a1392f29313caabcd99e3'),
  plot: 'A young French soldier in World War I is overcome with guilt when he kills a German soldier who, like himself, is a musically gifted conscript.',
  genres: [
    'Drama'
  ],
  runtime: 76,
  cast: [
    'Lionel Barrymore',
    'Nancy Carroll',
    'Phillips Holmes',
    'Louise Carter'
  ],
  num_mflix_comments: 2,
  poster: 'https://m.media-amazon.com/images/M/MV5BYzJlMDZjM2QtNWU0Yy00ZWGRilTLjNjc2ZGZjNGM2ZmIzN2I1XkEyXkFqcGdeQXVyMjUxODE0MDY@._V1_SV1000_SX677_AL_.jpg',
  title: 'Broken Lullaby',
  fullplot: "A young French soldier in World War I is overcome with guilt when he kills a German soldier who, like himself, is a musically gifted conscript.",
  languages: [
    'English'
  ],
  released: 1932-01-24T00:00:00.000Z,
  directors: [
    'Ernst Lubitsch'
  ],
  writers: [
    'Maurice Rostand (play)',
    'Reginald Berkeley (adaptation)'
  ]
}
```

```
{
  year: 1932,
  $or: [
    { genres: "Drama" },
    { languages: "French" }
  ]
}
```

[Generate query](#)
[Explain](#)
[Reset](#)
[Find](#)
[Options](#)

+ ADD DATA

UPDATE

DELETE

EXPORT DATA

EXPORT CODE

25

1 - 18 of 18

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```
_id: ObjectId('573a1391f2913caabcd9458')
plot: "A young artist draws a face at a canvas on his easel. Suddenly the mou..."
runtime: 55
rated: "UNRATED"
cast: Array (4)
num_mflix_comments: 1
poster: "https://m.media-amazon.com/images/M/MV5BYWY3ODE5ZWEtYjlmYi00NjA4LTk4ZW..."
title: "The Blood of a Poet"
lastupdated: "2015-09-16 13:13:05.537000000"
languages: Array (1)
released: 2010-05-20T00:00:00.000+00:00
directors: Array (1)
writers: Array (1)
awards: Object
year: 1932
imdb: Object
countries: Array (1)
type: "movie"
tomatoes: Object
```

```
_id: ObjectId('573a1392f2913caabcd99a3')
plot: "Junta is hated by the people in the village where she lives, especiall..."
genres: Array (3)
runtime: 85
cast: Array (4)
```

Vista del Compass, con el mismo query para ver cuántos resultados hay.

3. Muéstrame todos los documentos de películas estadounidenses que tengan entre 5 y 9 premios que fueron producidas entre 2012 y 2014.

```
> db.movies.find({
  countries: "USA",
  "awards.wins": { $gte: 5, $lte: 9 },
  year: { $gte: 2012, $lte: 2014 }
})
< {
  _id: ObjectId('573a13acf29313caabd29366'),
  fullplot: "The manager of the negative assets sector of Life magazine, Walter Mitty, has been working for sixteen years for the magazine and has
  imdb: {
    rating: 7.4,
    votes: 211230,
    id: 359950
  },
  year: 2013,
  plot: 'When his job along with that of his co-worker are threatened, Walter takes action in the real world embarking on a global journey that tur
  genres: [
    'Adventure',
    'Comedy',
    'Drama'
  ],
  rated: 'PG',
  metacritic: 54,
  title: 'The Secret Life of Walter Mitty',
  lastupdated: '2015-08-31 00:10:51.747000000',
  languages: [
    'English',
    'Spanish',
    'Icelandic'
  ],
}
```

Generate query **Explain** **Reset** **Find** **Options**

```
{
  countries: "USA",
  "awards.wins": { $gte: 5, $lte: 9 },
  year: { $gte: 2012, $lte: 2014 }
}
```

+ ADD DATA **UPDATE** **DELETE** **EXPORT DATA** **EXPORT CODE**

25 1 - 25 of 166

_id: ObjectId('573a13acf29313caabd29366')

fullplot: "The manager of the negative assets sector of Life magazine, Walter Mit..."

> imdb: Object

year: 2013

plot: "When his job along with that of his co-worker are threatened, Walter t..."

> genres: Array (3)

rated: "PG"

metacritic: 54

title: "The Secret Life of Walter Mitty"

lastupdated: "2015-08-31 00:10:51.747000000"

> languages: Array (3)

> writers: Array (3)

type: "movie"

> tomatoes: Object

poster: "https://m.media-amazon.com/images/M/MV5BODYwNDYxNDk1Nl5BMl5BanBnXkFtZT..."

num_mflix_comments: 1

released: 2013-12-25T00:00:00+00:00

☒ **awards:** Object

wins: 6

nominations: 13

text: "6 wins & 13 nominations."

> countries: Array (2)

0: "USA"

1: "Canada"

> cast: Array (4)

> directors: Array (1)

runtime: 114

Un poco más claro con esta vista.

NIVEL 2:

1. Cuenta cuántos comentarios escribe un usuario que utiliza "GAMEOFTHRON.ES" como dominio de correo electrónico.

```
> db.comments.countDocuments({  
  email: /gameofthron\.es$/i  
})  
< 22841
```

Los "/" son para buscar esa palabra o caracteres exactos, se usa la "\" para separar el "." porque si no se señala, tiene otro significado para el compilador. El "\$" es para decir que se busque al final del texto y la "i" hace que la búsqueda sea insensible a mayúsculas o minúsculas.

2. ¿Cuántos cines existen en cada código postal situados dentro del estado Washington DC (DC)?

```

> db.theaters.aggregate([
  {
    $match: {
      "location.address.state": "DC"
    }
  },
  {
    $group: {
      _id: "$location.address.zipcode",
      total_theaters: { $sum: 1 }
    }
  }
])
< {
  _id: '20002',
  total_theaters: 1
}
{
  _id: '20010',
  total_theaters: 1
}
{
  _id: '20016',
  total_theaters: 1
}

```

NIVEL 3:

1. Encuentra todas las películas dirigidas por John Landis con una puntuación IMDb (Internet Movie Database) de entre 7,5 y 8.

```

> db.movies.find({
  directors: "John Landis",
  "imdb.rating": {
    $gte: 7.5,
    $lte: 8
  }
})
< {
  _id: ObjectId('573a1397f29313caabce6d94'),
  fullplot: "Faber College has one frat house so disreputable it will take anyone. It has a second one full of white, anglo-saxon, rich young men w
  imdb: {
    rating: 7.6,
    votes: 84834,
    id: 77975
  },
  year: 1978,
  plot: 'At a 1962 college, Dean Vernon Wormer is determined to expel the entire Delta Tau Chi Fraternity, but those trouble-makers have other plan
  genres: [
    'Comedy'
  ],
  rated: 'R',
  metacritic: 82,
  title: 'Animal House',
  lastupdated: '2015-09-13 00:02:47.803000000',
  languages: [
    'English',
    'Italian'
  ],
  writers: [
    'Harold Ramis',
    'Douglas Kenney',
    'Chris Miller'
  ]
}

```

2. Muestra en un mapa la ubicación de todos los teatros de la base de datos.

```
> db.theaters.find(
  {},
  {
    theaterId: 1,
    "location.geo": 1
  }
)
< {
  _id: ObjectId('59a47286cfa9a3a73e51e72c'),
  theaterId: 1000,
  location: {
    geo: {
      type: 'Point',
      coordinates: [
        -93.24565,
        44.85466
      ]
    }
  }
}
{
  _id: ObjectId('59a47286cfa9a3a73e51e72d'),
  theaterId: 1003,
  location: {
    geo: {
      type: 'Point',
      coordinates: [
        -76.512016,
        38.29697
      ]
    }
  }
}
```

Este código me devuelve las coordenadas geospaciales [longitud, latitud] de cada cine.

{ "location.geo.type": "Point" }

Generate query

Explain

ADD DATA

UPDATE

DELETE

EXPORT DATA

EXPORT CODE

251 - 25 of 1564

_id: ObjectId('59a47286cfa9a3a73e51e72c')

theaterId: 1000

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e72d')

theaterId: 1003

location: Object

address: Object

geo: Object

type: "Point"

coordinates: Array (2)

0: -76.512016

1: 38.29697

_id: ObjectId('59a47286cfa9a3a73e51e72e')

theaterId: 1008

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e72f')

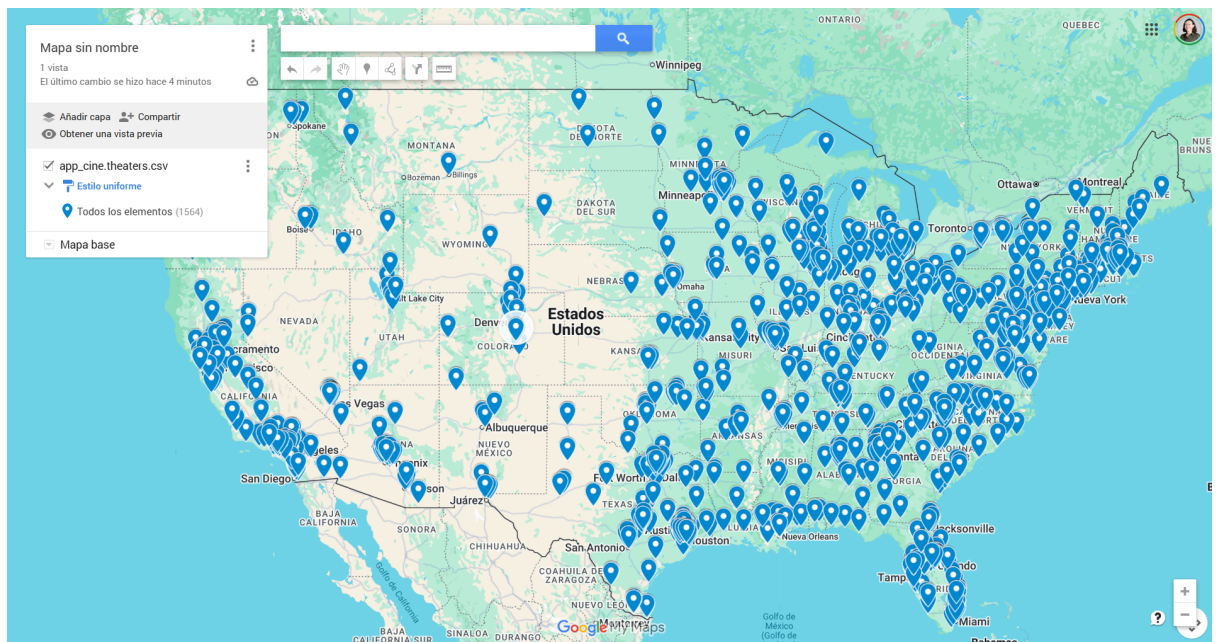
theaterId: 1004

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e730')

theaterId: 1002

location: Object



Al exportar el resultado de la query en un archivo CSV, éste se puede abrir en Google My Maps para visualizar las distintas ubicaciones.